

MATH 1610 / HONORS CALCULUS

Honors Calculus 1610

Lecture 6: Properties of the Integral, and Limits

September 21, 2026

Puzzler

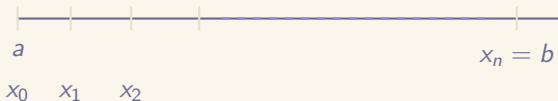
If you list the first one million natural numbers in **alphabetical order**, what would be the first **odd** number on the list?

(Herb Wilf, former Penn professor)

Last time: defining integrals via step functions

If s is a step function on $[a, b]$, then $\int_a^b s(x) dx$ is what you get by adding up the rectangles:

$$\int_a^b s(x) dx = \sum_{i=1}^n c_i (x_i - x_{i-1}).$$



Lower and upper integrals

For a function f on $[a, b]$:

$$L = \sup \left\{ \int_a^b s(x) \, dx \mid s \text{ a step function, } s(x) \leq f(x) \right\},$$

$$U = \inf \left\{ \int_a^b t(x) \, dx \mid t \text{ a step function, } t(x) \geq f(x) \right\}.$$

Definition

If $L = U$ we say f is **integrable**, and $\int_a^b f(x) \, dx = L = U$.

Properties of integrals

$$\int_a^b (f(x) + g(x)) \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx$$

$$\int_a^b \lambda f(x) \, dx = \lambda \int_a^b f(x) \, dx$$

Today: why the first one is true.

Additivity over the interval

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx.$$

$$a < b < c \quad \longleftarrow \quad \text{ignore this}$$

We do *not* want to assume any ordering of a , b , c . To get away with that, we need a convention.

The convention

Taking $c = a$ in the additivity property,

$$\int_a^b f(x) dx + \int_b^a f(x) dx = \int_a^a f(x) dx = 0.$$

So it is forced, and consistent, to define

$$\boxed{\int_b^a f(x) dx = - \int_a^b f(x) dx}$$

and then additivity holds for **any** ordering of the endpoints.

Check: the sum of two step functions is a step function



Step 1

Find a common set of endpoints $x_0, \dots, x_n$ that works for **both** f and g .

Then $f(x) = c_i$ and $g(x) = d_i$ on (x_{i-1}, x_i) , so $f(x) + g(x) = c_i + d_i$ there.

And the integral adds

$$\begin{aligned}\int_a^b (f(x) + g(x)) \, dx &= \sum_{i=1}^n (x_i - x_{i-1})(c_i + d_i) \\&= \sum_{i=1}^n \left((x_i - x_{i-1})c_i + (x_i - x_{i-1})d_i \right) \\&= \sum_{i=1}^n (x_i - x_{i-1})c_i + \sum_{i=1}^n (x_i - x_{i-1})d_i \\&= \int_a^b f(x) \, dx + \int_a^b g(x) \, dx.\end{aligned}$$

Now for integrable functions

Suppose f and g are integrable. We want to show

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx.$$

Give the competitors names:

$$S_f = \{s \text{ a step function} \mid s(x) \leq f(x)\},$$

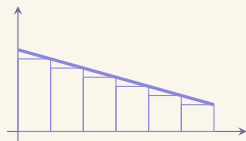
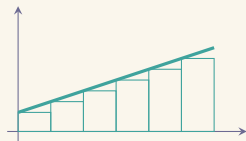
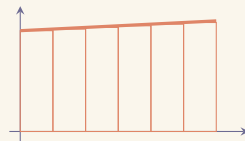
$$T_f = \{t \text{ a step function} \mid t(x) \geq f(x)\},$$

and likewise S_g, T_g . Write L_f, U_f, L_g, U_g for the corresponding sups and infs.

Adding the competitors

$$S = \{s_1 + s_2 \mid s_1 \in S_f, s_2 \in S_g\}, \quad T = \{t_1 + t_2 \mid t_1 \in T_f, t_2 \in T_g\},$$

$$\tilde{L} = \sup \left\{ \int_a^b s \mid s \in S \right\}, \quad \tilde{U} = \inf \left\{ \int_a^b t \mid t \in T \right\}.$$

 f  g  $f + g$

What we actually want

$$L = \sup \left\{ \int_a^b s(x) \, dx \mid s \leq f + g \right\}, \quad U = \inf \left\{ \int_a^b t(x) \, dx \mid t \geq f + g \right\},$$

and $\int_a^b (f + g) = L = U$ if these agree.

Hope

$$L = \tilde{L}, \quad U = \tilde{U}.$$

Careful: *not* every step function below $f + g$ has the form $s_1 + s_2$ — so this is a hope, not an observation.

Recollection

If $S_1, S_2 \subseteq \mathbb{R}$ are nonempty and bounded above, then

$$\sup \{a + b \mid a \in S_1, b \in S_2\} = \sup(S_1) + \sup(S_2).$$

And the same for infima: if S_1, S_2 are bounded below, then

$$\inf \{a + b \mid a \in S_1, b \in S_2\} = \inf(S_1) + \inf(S_2).$$

This is what turns $\tilde{L}$ and $\tilde{U}$ into something we recognize.

Computing $\widetilde{U}$

$$\begin{aligned}\widetilde{U} &= \inf \left\{ \int_a^b t(x) \, dx \mid t \in T \right\} \\ &= \inf \left\{ \int_a^b (t_1 + t_2) \mid t_1 \in T_f, t_2 \in T_g \right\} \\ &= \inf \left\{ \int_a^b t_1 + \int_a^b t_2 \mid t_1 \in T_f, t_2 \in T_g \right\} \\ &= \inf \left\{ \int_a^b t_1 \mid t_1 \in T_f \right\} + \inf \left\{ \int_a^b t_2 \mid t_2 \in T_g \right\} \\ &= U_f + U_g,\end{aligned}$$

The squeeze

The same computation gives $\tilde{L} = L_f + L_g$.

Every element of S is a step function below $f + g$, and every element of T is one above it. So

$$\tilde{L} \leq L \leq U \leq \tilde{U}.$$

But f and g are integrable, so $L_f = U_f$ and $L_g = U_g$, and therefore

$$\tilde{L} = L_f + L_g = U_f + U_g = \tilde{U}.$$

The two ends agree, so everything in between does: $L = U = \int_a^b f + \int_a^b g$. $\square$

Limits

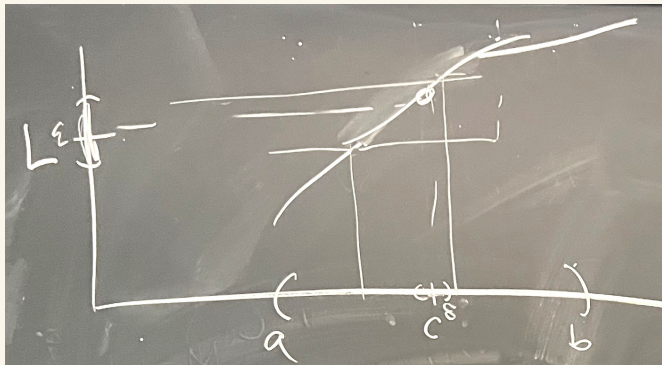
Suppose $f: (a, b) \rightarrow \mathbb{R}$ is a function and $c \in (a, b)$.

We say that

$$\lim_{x \rightarrow c} f(x) = L$$

to mean that we can ensure $f(x)$ is as close as we want to L by making x sufficiently close to c .

The picture



ϵ measures closeness to L ; δ measures closeness to c .

Formal

Definition

$\lim_{x \rightarrow c} f(x) = L$ means: for all real numbers $\epsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{whenever} \quad 0 < |x - c| < \delta.$$

The adversary picks ϵ ; we answer with δ . We get to see ϵ first.

Correction. On the board in class the $0 <$ was missing. It matters: the definition deliberately says nothing about the value $f(c)$, and indeed f need not be defined at c at all. Without the $0 <$, “the limit exists at c ” would already force $f(c) = L$, and the definition of continuity would have nothing left to say. This is the convention in Apostol, §3.2.

Example: a constant function

If $f(x) = M$ for all x , then $\lim_{x \rightarrow c} f(x) = M$ for every c .

Given $\epsilon > 0$ — how do we find $\delta > 0$ so that ...?

Choose any $\delta > 0$. Then if $|x - c| < \delta$,

$$|f(x) - M| = |M - M| = 0 < \epsilon. \quad \checkmark$$

Example: the identity function

If $f(x) = x$, then $\lim_{x \rightarrow c} x = c$.

Given $\epsilon > 0$, choose $\delta = \epsilon$. Then

$$|x - c| < \epsilon \implies \underbrace{|f(x) - c|}_{=x} = |x - c| < \epsilon. \quad \checkmark$$

Worksheet 3

Show that

$$\underbrace{\lim_{x \rightarrow a} f(x) = L}_{(1)} \iff \underbrace{\lim_{h \rightarrow 0} f(a + h) = L}_{(2)}$$

Unwinding the two statements:

$$(1): \forall \epsilon > 0 \exists \delta > 0 \text{ st } |x - a| < \delta \implies |f(x) - L| < \epsilon,$$

$$(2): \forall \epsilon > 0 \exists \delta > 0 \text{ st } |h - 0| < \delta \implies |f(a + h) - L| < \epsilon.$$

The variable is a dummy. The substitution to make is $\boxed{h = x - a}$, or $x = h + a$.

Assume (2), show (1)

Given $\epsilon > 0$, we want a δ .

By (2) we can find δ so that whenever $|h| < \delta$ we have $|f(a + h) - L| < \epsilon$.

Let $h = x - a$. Then

$$\begin{aligned} |x - a| < \delta &\implies |h| < \delta \implies |f(a + h) - L| < \epsilon \\ &\implies |f(a + (x - a)) - L| < \epsilon \implies |f(x) - L| < \epsilon. \quad \square \end{aligned}$$

The other direction is the same substitution, read backwards.

Doing one by hand

$$\lim_{x \rightarrow 3} x^2 = 9.$$

Suppose $\epsilon > 0$. We want $|x^2 - 9| < \epsilon$, that is

$$|x - 3| |x + 3| < \epsilon.$$

The second factor needs controlling first:

$$0 \leq |x - 3| < 1 \implies |x + 3| < 7.$$

Doing one by hand, concluded

Let

$$\delta = \min \left\{ \frac{\epsilon}{7}, 1 \right\}.$$

Then if $|x - 3| < \delta$,

$$|x^2 - 9| = |x - 3| |x + 3| \leq 7|x - 3| < 7 \cdot \frac{\epsilon}{7} = \epsilon. \quad \square$$

Notice how much of the work went into a bound that has nothing to do with the answer. We do not want to do this for every polynomial — next time we build up **continuity** instead.